Chapter 25 : Robotics

1. What is Robotics?

- Robotics is the study and design of autonomous or semi-autonomous physical agents that sense, plan, and act in the real world.
- Combines AI, control, perception, mechanics, and hardware.
- Robots must operate under uncertainty, changing environments, and physical constraints.
- Goal: achieve useful tasks safely, reliably, and efficiently.

Key Idea: A robot is an embodied agent that perceives, reasons, and acts in the physical world.

2. What Makes a Robot Intelligent?

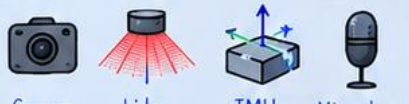
- Perceives the environment using sensors.
- Maintains an internal model or belief about the world.
- Plans actions to achieve goals.
- Adapts to uncertainty and dynamic changes.
- Learns from experience (optional).
- Interacts safely and effectively with people and objects.

Example: A vacuum cleaning robot builds a map, plans a cleaning route, avoids obstacles, and returns to charge when the battery is low.

3. Sensors and Actuators

- Sensors measure properties of the environment and the robot's internal state.
- Actuators change the robot's state or the environment.
- Many robots use multiple sensors and actuators working together.

Common Sensors:

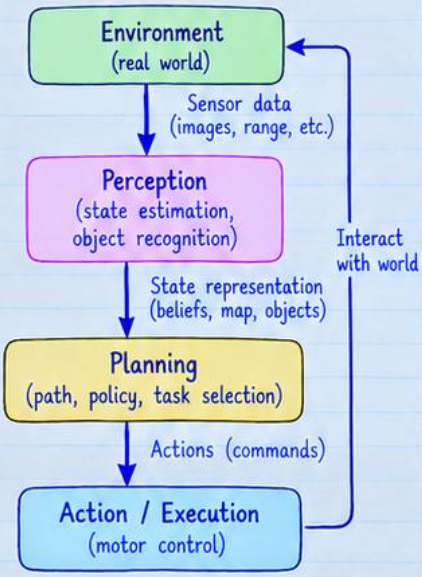


Common Actuators:



4. Perception–Planning–Action Loop

- Robots continuously sense, interpret, plan, and act in a loop.
- The loop allows them to react to changes and achieve long-term goals.
- Perception converts raw sensor data into a useful state representation.
- Planning selects actions using search, optimization, or learned policies.
- Execution sends commands to actuators and the cycle repeats.



5. Robot Environments

- Robots operate in diverse, complex, and dynamic environments.
- Environments may be partially observable, stochastic, and continuous.
- Robots must handle moving objects, changing layouts, and uncertainty.
- Examples: indoor homes, factories, outdoor urban areas, underwater, space, and aerial environments.

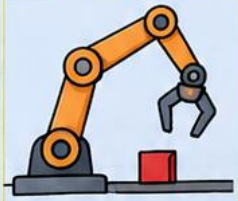




6. Manipulators vs. Mobile Robots

- Manipulators are fixed or arm-like robots that interact with objects using end-effectors (e.g., grippers, tools).
- Mobile robots move through the environment using wheels, legs, tracks, or flying mechanisms.
- Many robots combine both (e.g., a mobile base with a robotic arm).

Manipulator (Robotic Arm)



- Fixed base (or on a mobile base)
- High-precision manipulation
- Used in assembly, pick-and-place, and surgical robots

Mobile Robot



- Moves in the environment
- Navigation and obstacle avoidance
- Used in delivery, exploration, cleaning, and surveillance

7. Autonomy in Robotics

- Autonomous robots perceive, plan, and act with minimal human intervention.
- Levels of autonomy range from teleoperation to full autonomy.
- Robots must handle uncertainty, failures, and unexpected situations.
- Human-robot interaction is important for shared control and safety.

Levels of Autonomy:

Less autonomous ⟷ More autonomous

Teleoperation (human controls actions)	Shared control (human + robot)	Supervised autonomy (human monitors)	Full autonomy (robot decides)

8. Real-World Applications

- Industrial automation: manufacturing, welding, assembly, quality inspection.
- Service robots: cleaning, delivery, agriculture.
- Exploration: planetary rovers, underwater robots, disaster response.
- Healthcare: surgical robots, rehabilitation robots, assistive robots.
- Logistics: warehouse robots, autonomous vehicles, drones.
- Personal and home robots: vacuum cleaners, companionship robots.

Example: A warehouse robot receives an order, navigates to the shelf, picks the item with a gripper, and delivers it to the packing station.

> "Robotics brings artificial intelligence into the real world, turning decisions into physical action." — AIMA (Ch. 25)

Chapter 25 : Robotics

1. What is Robotics?

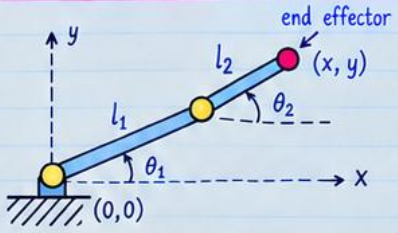
- Robotics deals with the design, control, and use of physical agents that act in the real world.
- Combines perception, planning, control, mechanics, and sensing.
- Robots must operate under uncertainty and interact with continuous environments.
- Key challenge: bridge the gap between symbolic decision making and continuous physical action.

Key Idea: A robot is an embodied agent that senses, plans, and acts in the physical world.

2. Robot Kinematics

- Kinematics describes the motion of a robot in terms of its geometry, without considering forces.
- A robot is a set of rigid links connected by joints.
- Joint variables (e.g., angles or translations) determine the configuration of the robot.
- Forward kinematics: compute the position and orientation of the end effector from joint variables.

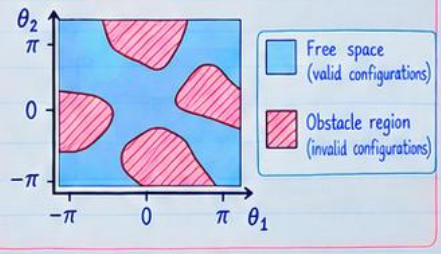
Example: Planar 2-link arm



3. Configuration Space

- The configuration space (C-space) is the set of all possible configurations of a robot.
- Each point in C-space represents a unique pose (position and orientation) of the robot.
- Obstacles in the real world map to forbidden regions in C-space.

Example: 2-link planar arm



Planning in C-space reduces a geometric motion problem to a path-finding problem.

4. Degrees of Freedom (DOF)

- Degrees of freedom (DOF) is the number of independent parameters needed to specify a robot's configuration.
- Common joints:
  - Revolute joint: 1 rotational DOF (angle)
  - Prismatic joint: 1 translational DOF (distance)
  - Spherical joint: 3 rotational DOF
- Example: A planar 2-link arm has 2 DOF (θ_1, θ_2).

Key Point: More DOF gives greater flexibility but increases the complexity of planning and control.

5. Effectors and End Effectors

- Effectors are the parts of the robot that interact with the environment.
- End effectors are the final link, such as:
  - Grippers (parallel jaw, vacuum, soft gripper)
  - Tools (e.g., drill, welder, screwdriver)
  - Sensors (e.g., camera, force sensor)

Examples:



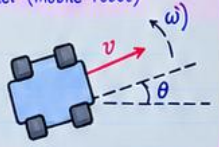
The end effector defines how the robot can affect the world (grasp, push, move, etc.).

6. Motion Models

- A motion model describes how the robot state changes over time.
- Types of motion models:
  - Kinematic model: relates control inputs to the next state (e.g., joint velocities).
  - Dynamic model: includes forces, torques, mass, and inertia (more accurate).
- Discrete-time model: $x_{t+1} = f(x_t, u_t)$ where x_t is the state and u_t is the control.
- Continuous-time model: $\dot{x} = f(x, u)$

Example: Unicycle model (mobile robot)

$$\dot{x} = v\cos\theta$$
$$\dot{y} = v\sin\theta$$
$$\dot{\theta} = \omega$$



7. Control in Robotics

- Control converts desired motions into motor commands.
- Common control approaches:
  - Open-loop control: execute a fixed sequence of commands.
  - Feedback control: use sensor information to correct errors (e.g., PID control).
  - Trajectory tracking: follow a desired path in joint or Cartesian space.
- Control must handle noise, delays, and model inaccuracies.

Key Idea: Feedback control uses sensor measurements to reduce error between desired and actual motion.

8. Inverse Kinematics (IK)

- Inverse kinematics computes the joint values needed to achieve a desired end-effector pose.
- May have multiple solutions, no solution, or be redundant.
- Analytic solutions exist for simple robots; otherwise iterative numerical methods are used (e.g., Jacobian transpose, gradient descent).

Example: 2-link planar arm IK

Given desired (x, y), find (θ_1, θ_2).

$$\cos\theta_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2 l_1 l_2}$$

$$\theta_2 = \pm \arccos(\cos\theta_2)$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(l_2 \sin\theta_2,\ l_1 + l_2\cos\theta_2)$$

9. Constraints and Feasibility

- Robots are subject to constraints:
  - Joint limits (e.g., $\theta_{\min} \le \theta \le \theta_{\max}$)
  - Collision avoidance with obstacles
  - Kinematic constraints (e.g., closed chains)
  - Dynamic constraints (e.g., torque limits)
- Feasible motions must satisfy all constraints.
- Planning and control must ensure safety and avoid self-collisions.

Example: Joint limits $\theta_1 \in [-\pi/2, \pi/2]$, $\theta_2 \in [-\pi/2, \pi/2]$

10. Key Takeaways

1. Kinematics describes geometry of motion.
2. Configuration space represents all possible states.
3. DOF determines robot flexibility.
4. Effectors enable physical interaction.
5. Motion models relate control inputs to state.
6. Feedback control is essential for robust motion.
7. Inverse kinematics finds joint values for a goal.
8. Constraints must be satisfied for feasible motion.
9. These concepts support planning and execution in real-world robotic systems.

"Robotics brings AI into the physical world, where perception, planning, and action must work together." — AIMA (Ch. 25)

Chapter 25 : Robotics

1. What is Robotics?

- Robotics deals with agents that operate in the physical world using sensors and actuators.
- Main problems: localization, mapping, motion planning, and control.
- Robots must perceive a noisy world, maintain an estimate of their state, and act to achieve goals.

Key Idea: A robot continuously estimates its state (position and orientation) while building and using a map of the environment.

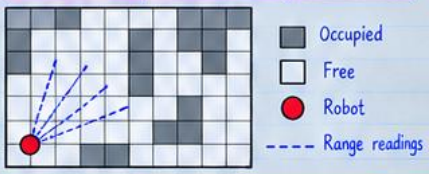
2. Localization

- Localization is the problem of estimating the robot's pose (position and orientation) in a known environment.
- Uses sensor measurements (e.g., range, bearing) and a motion model.
- Often formulated as a probabilistic state estimation problem.
- Types:
  - Global localization: find pose anywhere in the map.
  - Local localization: track pose from a known initial position.

3. Mapping

- Mapping is the problem of constructing a map of an unknown environment.
- Uses sensor data collected as the robot moves (e.g., lidar, camera, sonar).
- Map representations:
  - Grid maps (occupancy grids)
  - Feature maps (landmarks)
  - Topological maps (graph of places)

Example: Build an occupancy grid from range sensor readings.



4. Uncertainty in Robot State Estimation

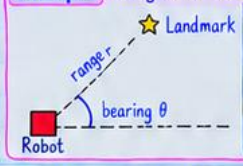
- Sensor measurements are noisy and imprecise (e.g., due to noise, occlusion, or limited resolution).
- Robot motion is also uncertain (e.g., wheel slip).
- Therefore, the robot maintains a probability distribution over possible states (e.g., position).
- Common representations: Gaussian (in Kalman Filter) or a set of weighted samples (in Particle Filter).

Example: A range sensor may return a noisy distance measurement, leading to uncertainty in the robot's position.

5. Landmarks and Sensor Models

- Landmarks are distinctive, reliably detectable features in the environment.
- Examples: corners, beacons, doors, visual features.
- Sensor model: gives the probability of a measurement given the state and a landmark.
- Helps correlate measurements with map features and reduce ambiguity.

Example: Range and bearing to a landmark



Measurement model:
$z = (r, \theta) = h(x, m) + \text{noise}$

where x is robot state, m is landmark position.

6. Dead Reckoning (Odometry)

- Estimates the new state by integrating the robot's own motion commands and wheel encoders.
- Uses a motion model: $x_t = f(x_{t-1}, u_t)$ + motion noise
- Simple and works over short distances.
- Errors accumulate over time due to noise and unmodeled effects (e.g., wheel slip).

Example: If a robot moves forward 1 m with some noise, the estimated position may have a Gaussian uncertainty around the expected location.

7. Kalman Filtering (Linear Gaussian Case)

- Kalman filter provides an efficient solution for linear systems with Gaussian noise.
- Maintains a Gaussian belief: mean μ and covariance Σ.
- Two steps: Prediction (motion update) and Correction (measurement update).
- Works well for small continuous state spaces (e.g., 2D/3D position and heading).

Equations (discrete time):

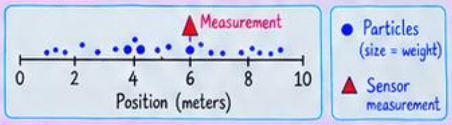
Prediction:	Correction:
$\bar{\mu}_t = A\mu_{t-1} + Bu_t$	$K_t = \bar{\Sigma}_t H^T (H\bar{\Sigma}_t H^T + Q)^{-1}$
$\bar{\Sigma}_t = A\Sigma_{t-1}A^T + R$	$\mu_t = \bar{\mu}_t + K_t(z_t - H\bar{\mu}_t)$
	$\Sigma_t = (I - K_t H)\bar{\Sigma}_t$

A, B: motion model H: sensor model
R: motion noise Q: measurement noise

8. Particle Filtering (Nonlinear / Non-Gaussian)

- Represents the belief as a set of weighted particles (samples).
- Works with nonlinear motion and sensor models.
- Algorithm:
  1. Initialize a set of particles.
  2. For each time step:
     a) Sample new particles using the motion model.
     b) Weight particles by the likelihood of the measurement.
     c) Resample according to weights (more likely particles are kept).
  3. The particle set approximates the posterior distribution.

Example: 1D Localization with Particles

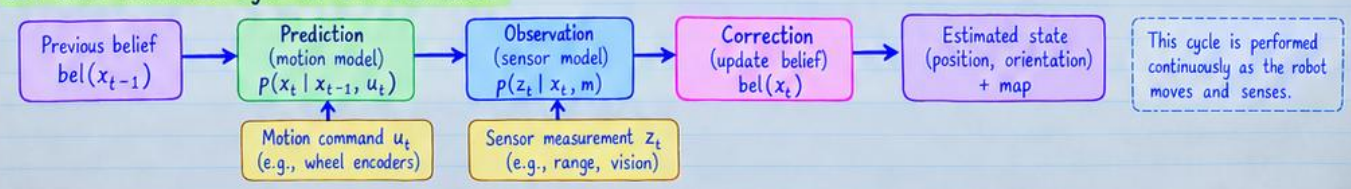


9. Simultaneous Localization and Mapping (SLAM)

- SLAM estimates the robot's pose and a map of the environment at the same time.
- Both the robot position and the map are uncertain and mutually dependent.
- Common approaches: EKF-SLAM, Particle Filter SLAM, Graph-based SLAM.
- Challenges:
  - Data association: which landmark was observed?
  - Computational complexity.
  - Loop closure: recognizing previously visited places.
- Applications: autonomous navigation, exploration, and mobile robotics.

Key Challenge: Correlating observations over time to build a consistent map while correcting the robot's position.

10. State Estimation Cycle for a Mobile Robot



Chapter 25 : Robotics

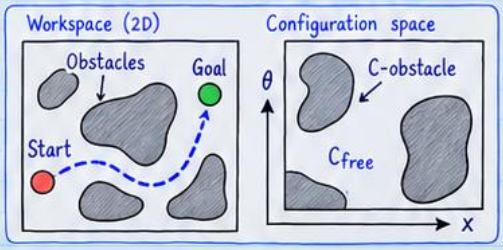
1. What is Motion Planning?

- Motion planning computes a sequence of robot configurations that moves the robot from a start to a goal while avoiding obstacles.
- Takes into account the robot's kinematics, geometry, and environmental constraints.
- Goal is a feasible, collision-free path, often optimized for length, time, or energy.

Key Idea: Find a continuous path in the configuration space from start to goal without collisions.

2. Configuration Space (C-Space)

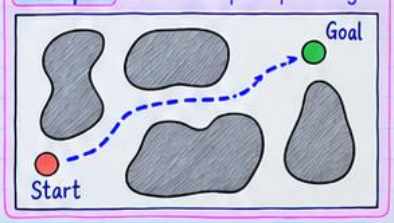
- A configuration specifies the position and orientation of the robot (all degrees of freedom).
- The configuration space (C-space) is the space of all possible configurations.
- Obstacles in the real world map to regions in C-space called C-obstacles.
- Planning reduces to finding a path from start to goal in the free space C_{free}.



3. Path Planning

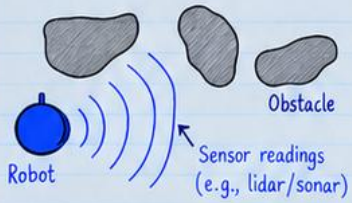
- Finds a collision-free path in C-space from start to goal.
- Can be posed as a graph search problem.
- Must account for robot shape, size, and kinematic constraints.
- Paths can be optimized (shortest, smooth, minimum time, etc.).

Example: 2D robot path planning



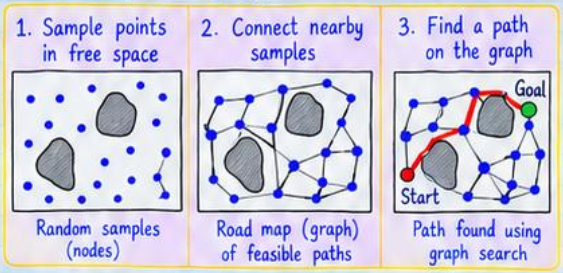
4. Obstacle Avoidance

- Robot must avoid collisions with static or moving obstacles.
- Uses sensing (e.g., cameras, lidar, sonar) and a planning or reactive method.
- Can be global (plan full path) or local (detect and avoid nearby obstacles).
- Must consider robot's size and shape (safety margin).



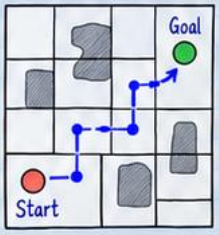
5. Road Maps (Probabilistic Roadmaps)

- Represent free space with a graph of collision-free configurations (nodes) and feasible paths (edges).
- Sample random configurations and connect nearby samples if the path is collision-free.
- Good for high-dimensional spaces.
- Probabilistic: completeness increases as more samples are added.



6. Cell Decomposition

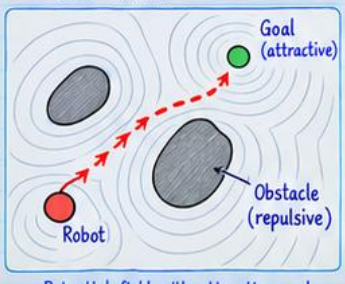
- Divide the free space into simple cells (e.g., grid cells, trapezoids, convex regions).
- Construct a graph where each cell is a node and edges connect adjacent free cells.
- Plan a path on the cell graph and then convert to a smooth path.



Types:
- Grid decomposition
- Trapezoidal decomposition
- Voronoi diagrams
- Convex decomposition

7. Potential Fields

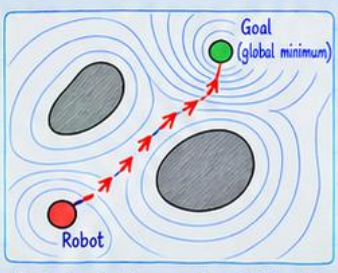
- Define an artificial potential function over the configuration space.
- Goal exerts an attractive potential; obstacles exert repulsive potentials.
- Robot follows the negative gradient of the potential (steepest descent).
- Simple and fast, but can get stuck in local minima.



Potential field with attractive and repulsive forces

8. Navigation Functions

- A special class of potential functions that guarantee convergence to the goal under certain conditions.
- Have no local minima except at the goal.
- Provide a well-defined gradient field for smooth navigation.
- Useful for robots with simple kinematics in known environments.

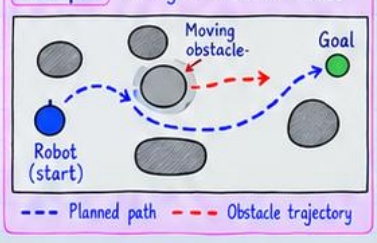


Navigation function: smooth potential with no local minima (except at goal)

9. Dynamic Environments

- Obstacles may move (e.g., people, vehicles).
- Requires sensing and continual replanning.
- Common approaches:
  - Predict motion of moving obstacles.
  - Use time-augmented configuration space (x, y, θ, t).
  - Replan locally when new obstacles are detected.
  - Use probabilistic or reactive methods (e.g., velocity obstacles, dynamic window).

Example: Moving obstacle avoidance



10. Navigation Examples

- Indoor robot navigation: mobile robots in offices or warehouses.
- Autonomous driving: plan routes on roads with static and dynamic obstacles.
- UAV (drone) navigation in cluttered environments.
- Planetary rover navigation on rough terrain.
- Manipulator motion planning in arms and grasping.

Example:
- Warehouse robot navigating to a goal while avoiding shelves and moving people.
- Uses a combination of global planning (road map) and local obstacle avoidance (potential fields).

Key Takeaways

1. Motion planning finds a collision-free path from start to goal.
2. Can use road maps, cell decomposition, potential fields, etc.
3. Potential fields are simple but can get stuck in local minima.
4. Navigation functions provide guaranteed convergence.
5. Dynamic environments require sensing and replanning.
6. Many real-world applications in robotics, autonomous driving, UAVs, and more.

> "Robotics brings AI into the physical world, where intelligent agents act, move, and interact with the environment." — AIMA (Ch. 25)

Chapter 25 : Robotics
Summary and Review

1. Manipulation and Grasping (Summary)

- Manipulation involves controlling a robot's end effector (e.g., gripper or hand) to interact with objects.
- Grasping plans the contacts, forces and finger motions to hold objects securely.
- Key issues: object shape, friction, force control, uncertainty, and robustness.
- Uses perception (vision/tactile) to locate and estimate object pose.

Key Idea: Manipulation requires both geometric reasoning and force control under uncertainty.

2. Planning and Acting in Robots

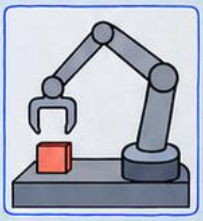
- Robots must sense, plan and act in dynamic, uncertain environments.
- Motion planning finds collision-free paths for the robot's body and manipulators.
- Task planning sequences high-level actions to achieve goals (e.g., pick and place).
- Execution and control convert plans into continuous, real-time actions.
- Must handle unexpected events (e.g., obstacles, slippage) using sensing and re-planning.

Key Idea: Integration of planning, perception and control enables robust real-world robot behavior.

3. Example: Pick-and-Place Pipeline

1. **Perceive** — Detect and localize object (e.g., with vision).
2. **Plan** — Choose grasp, plan collision-free arm motion.
3. **Control** — Execute motion with feedback (position / force control).
4. **Monitor** — Check success, handle failures and re-plan if needed.

Example: Pick and Place



Combines perception, motion planning, control and feedback to achieve the task robustly.

4. Types of Grasping

- Power grasp: strong, stable grasp for heavier objects; uses multiple contacts.
- Precision grasp: accurate control for smaller or delicate objects.
- Form closure: geometry alone prevents object motion.
- Force closure: combination of geometry and friction prevents motion.
- Grasp planning searches over grasps using models or learned approaches.

Example: A parallel-jaw gripper can perform a power grasp on a box, while a pinch grasp is used for a small object.

5. Applications of Robotics

- Industrial manufacturing: assembly, welding, painting, material handling.
- Healthcare: surgical robots, rehabilitation, assistive robots.
- Service robots: household tasks, delivery, cleaning.
- Exploration: space rovers, underwater robots.
- Agriculture: harvesting, monitoring.
- Autonomous vehicles: robots on wheels, air or water.

Application Examples:



Industrial robot (assembly) | Planetary rover (exploration) | Aerial robot (mapping)

6. Current Challenges

- Robust perception in complex, changing environments (lighting, clutter, occlusion).
- Reliable grasping of novel and deformable objects (e.g., clothes, food).
- Planning and control under uncertainty.
- Learning from raw sensor data and demonstration (e.g., imitation learning).
- Generalization to new tasks and environments.
- Safe and reliable operation around people.
- Combining symbolic planning with continuous control and learning.

Key Challenge: Achieving general, adaptive and safe manipulation and mobility in the real world.

7. Safety and Ethical Considerations

- Robots operate in close proximity to people.
- Must detect and avoid collisions.
- Use safety-critical control, force/torque sensing and compliant actuators.
- Fail-safe behaviors: stop, retreat, or switch to safe mode on anomaly.
- Ethical issues: privacy, accountability, job displacement, and military use.

Key Idea: Safe, fair and beneficial robotics requires technical, social and ethical solutions.

8. Comparison of Robotics Tasks

Task	Goal	Key Techniques	Example
Perception	Understand environment	Vision, Lidar, SLAM, sensor fusion	Object recognition
Manipulation	Interact with objects	Grasp planning, motion planning, force control	Pick and place
Navigation	Move in space	Mapping, localization, path planning	Mobile robot in building
Task Planning	Sequence actions to achieve goal	Heuristic search, planning, learning	Assemble product
Control	Execute motions	Feedback control, trajectory tracking	Move robotic arm

9. Key Terms (Review)

- Manipulator: robotic arm or hand.
- End effector: device that interacts with objects (e.g., gripper).
- Grasp: set of contacts and forces to hold an object.
- Form closure: object is immobilized by geometry.
- Motion planning: find collision-free paths.
- SLAM: simultaneous localization and mapping.
- Force control: regulate interaction forces.
- Task planning: sequence of high-level actions.
- Compliance: yielding to external forces.
- Human-robot interaction (HRI): robots working with people.

10. Exam-Style Revision Points

- Explain the robot manipulation pipeline (perception → planning → control → execution).
- Compare power grasp and precision grasp.
- Describe how motion planning avoids collisions.
- Discuss the role of perception in grasping.
- List and explain real-world applications of robotics.
- Outline current challenges in robotics.
- Explain safety considerations for human-robot interaction.
- Be able to define key terms: manipulator, end effector, grasp, SLAM, force control, task planning.
- Design a high-level plan for a pick-and-place task.

11. Chapter Key Takeaways

1. Robotics integrates perception, planning, control and learning to act in the real world.
2. Manipulation and grasping require geometric and force reasoning under uncertainty.
3. Robots must plan and execute actions while coping with dynamic environments.
4. Robotics has wide-ranging applications from industry to healthcare and exploration.
5. Key challenges include robust perception, reliable manipulation, generalization and safety.
6. Combining symbolic and sub-symbolic methods is essential for real-world robotics.
7. Robotics is a central testbed for intelligent action in AI.

12. Mini Table: Sensing for Robotics

Sensor	What it Measures	Typical Use
Camera	2D/3D visual information	Object detection, pose estimation
Lidar	3D distance measurements	Mapping, localization
Tactile sensor	Contact, pressure	Grasping, manipulation
Force/Torque sensor	Interaction forces and torques	Compliant control, safe interaction
IMU	Acceleration and angular rate	State estimation (odometry)
Prosimity sensor	Distance to objects	Collision avoidance

"Robotics turns intelligent decision making into physical action, bringing AI out into the real world." — AIMA (Ch. 25)